

HDOC-6(EXPERIMENT)

- Q.1-A square photo has side `s` centimeters. A border of width `w` centimeters is added around it. Write a C program to find the total area after adding the border.

ALGORITHM/ PSEUDOCODE:

1. Start
2. Declare the variables s, w, newside, area
3. read the values of s and w
4. Calculate the newside = $s + (2 \times w)$
5. Calculate area = newside x newside
6. Display the total area
7. STOP

Test case	Input(s, w)	Expected output	Actual output	Result
1.	20,5	900 sq.cm	900 sq.cm	Pass
2.	15,5	625 sq.cm	625 sq.cm	Pass
3.	10,2	196 sq.cm	196 sq.cm	Pass

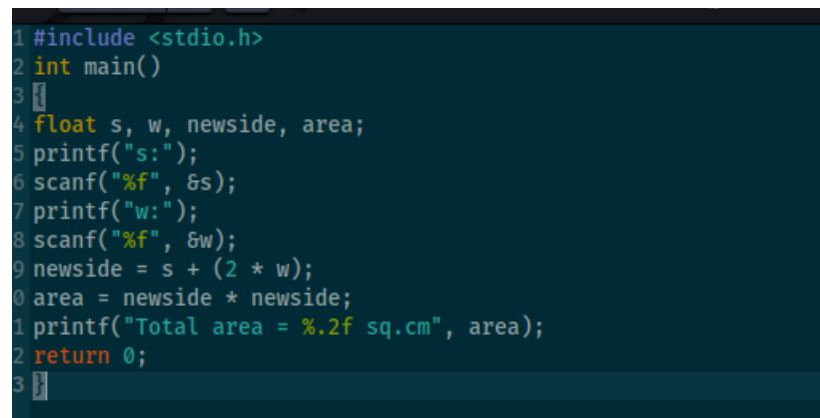
The code that we are going to use

```
#include <stdio.h>

int main()
{
    float s, w, newside, area;
    printf("s:");
    scanf("%f", &s);
    printf("w:");
    scanf("%f", &w);
    newside = s + (s * w);
    area = newside * newside;
    printf("Total area = %.2f sq.cm", area);
    return 0;
}
```

Execution of test cases

The source code

A screenshot of a code editor with a dark background and light-colored text. The code is written in C and is the same as the code block above. It includes a preprocessor directive for stdio.h, a main function, variable declarations for s, w, newside, and area, and a series of printf and scanf statements to get input and calculate the total area. The code is numbered from 1 to 23.

```
1 #include <stdio.h>
2 int main()
3 {
4     float s, w, newside, area;
5     printf("s:");
6     scanf("%f", &s);
7     printf("w:");
8     scanf("%f", &w);
9     newside = s + (2 * w);
10    area = newside * newside;
11    printf("Total area = %.2f sq.cm", area);
12    return 0;
13 }
```

Answer of the three cases

```

(kali㉿kali)-[~]
$ gedit hdoc.c

** (gedit:10457): WARNING **: 05:28:18.817:
** (gedit:10457): WARNING **: 05:28:18.817:

(kali㉿kali)-[~]
$ cc -g hdoc.c

(kali㉿kali)-[~]
$ ./a.out
s:20
w:5
Total area = 900.00 sq.cm

(kali㉿kali)-[~]
$ ./a.out
s:15
w:5
Total area = 625.00 sq.cm

(kali㉿kali)-[~]
$ ./a.out
s:10
w:2
Total area = 196.00 sq.cm

(kali㉿kali)-[~]
$ 

```

Now that we have made the algorithm the test case table and executed the code we can move onto the second question which is

- Q.2 Write a C program to find the fencing cost, area, and perimeter of a trapezium-shaped field when all sides, height, and fencing rate per meter are given.

Algorithm/Pseudocode

1. Start
2. Declare variables a, b, c, d, h, rate, area, perimeter, cose
3. Read the values of a, b, c, d, h and rate
4. Calculate area = $((a + b) \times h) / 2$
5. calculate perimeter = $a + b + c + d$
6. calculate cost = perimeter x rate
7. display area, perimeter and fencing cost
8. stop

Here:

a = first parallel side

b = second parallel side

c = side1

d = side 2

h = height

TEST CASE TABLE:

Test case	Input(a, b ,c, d, h, rate	Expected output	Actual output	Result
1	10, 8, 6, 9, 6, 110	Area = 54 sq.cm, perimetr = 33m, fencing cost = 3630	same	Pass

2	12, 7, 8, 10, 7, 150	Area = 66.5 sq.cm, perimeter = 37m, fencing cost = 5550	same	Pass
3	14, 9, 10, 12, 9, 200	Area = 103.5 sq.cm Perimeter = 45 m, fencing cost = 9000	same	pass

The code we are going to use

```
#include <stdio.h>

Int main()
{
float a, b, c, d, h, rate;
float area, perimeter, cost
print("a:");
scanf("%f", &a);
printf("b:");
scanf("%f", &b);
printf("c:");
scanf("%f", &c);
printf("d:");
scanf("%F", &d);
printf("h:");
scanf("%f", &h);
printf("rate:");
```

```
scanf("%f", &rate);  
area = ((a + b) * h) / 2;  
perimeter = a + b + c + d;  
cost = perimeter * rate;  
printf("Area = %.2f sq.m", area);  
printf("perimeter = %.2f m", perimeter);  
printf("fencing cost = Rs. %.2f", cost);  
return 0;  
}
```

Execution of the three test given in the table

```
1 #include <stdio.h>
2
3 int main()
4 {
5     float a, b, c, d, h, rate;
6     float area, perimeter, cost;
7
8     printf("a: ");
9     scanf("%f", &a);
10
11    printf("b: ");
12    scanf("%f", &b);
13
14    printf("c: ");
15    scanf("%f", &c);
16
17    printf("d: ");
18    scanf("%f", &d);
19
20    printf("h: ");
21    scanf("%f", &h);
22
23    printf("rate: ");
24    scanf("%f", &rate);
25
26    area = ((a + b) * h) / 2;
27    perimeter = a + b + c + d;
28    cost = perimeter * rate;
29
30    printf("Area = %.2f sq.m", area);
31    printf("Perimeter = %.2f m", perimeter);
32    printf("Fencing Cost = Rs. %.2f", cost);
33
34    return 0;
```

```
(kali㉿kali)-[~]
$ gedit hdoc2.c

** (gedit:34646): WARNING **: 06:16:59.290: Could not load Peas rep
** (gedit:34646): WARNING **: 06:16:59.290: Could not load PeasGtk

(kali㉿kali)-[~]
$ cc -g hdoc2.c

(kali㉿kali)-[~]
$ ./a.out
a: 10
b: 8
c: 6
d: 9
h: 6
rate: 110
Area = 54.00 sq.mPerimeter = 33.00 mFencing Cost = Rs. 3630.00

(kali㉿kali)-[~]
$ ./a.out
a: 12
b: 7
c: 8
d: 10
h: 7
rate: 150
Area = 66.50 sq.mPerimeter = 37.00 mFencing Cost = Rs. 5550.00

(kali㉿kali)-[~]
$ ./a.out
a: 14
b: 9
c: 10
d: 12
h: 9
rate: 200
Area = 103.50 sq.mPerimeter = 45.00 mFencing Cost = Rs. 9000.00

(kali㉿kali)-[~]
$ gedit hdoc2.c

** (gedit:37915): WARNING **: 06:22:47.280: Could not load Peas rep
** (gedit:37915): WARNING **: 06:22:47.280: Could not load PeasGtk

(kali㉿kali)-[~]
$
```